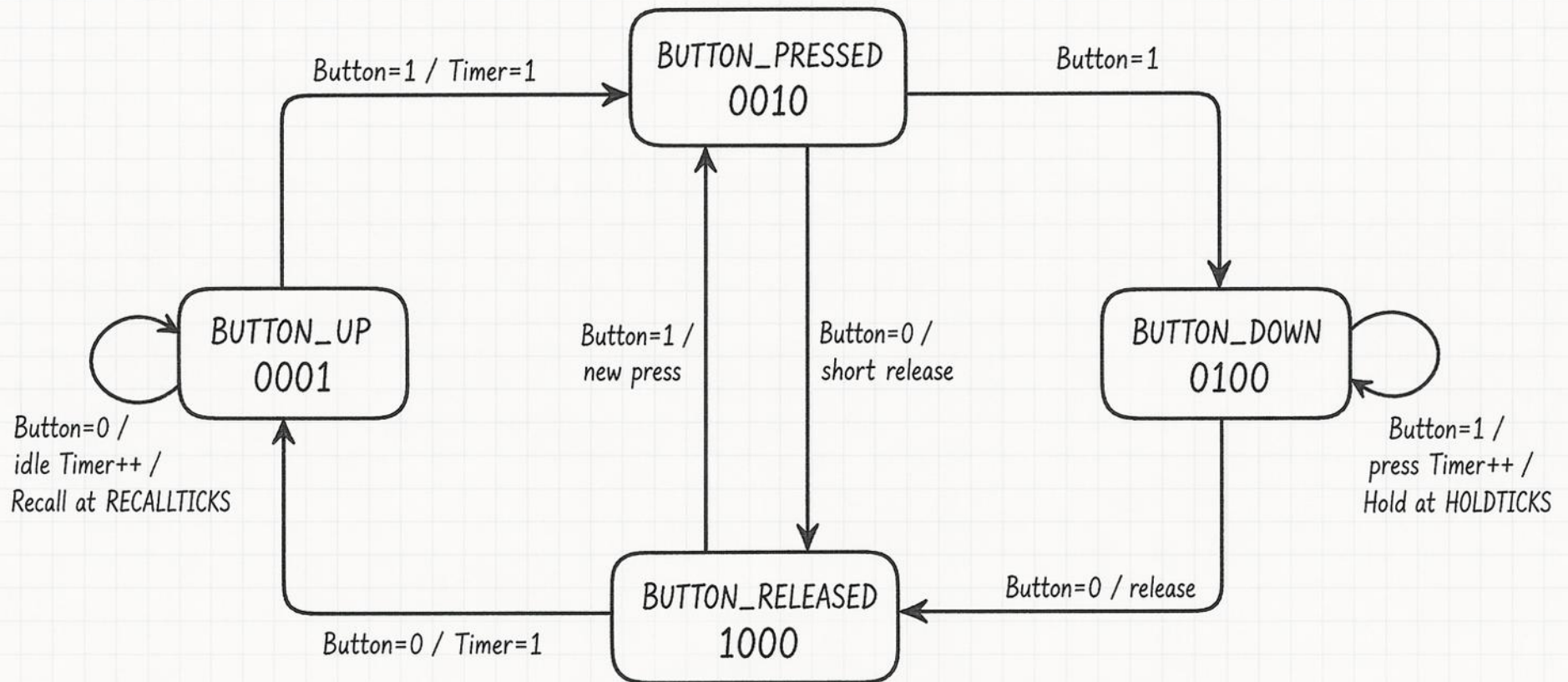


Headlamp Button Controller FSM

FSM diagram and waveform plot generated from the implemented design and real simulation data.

Headlamp Button Controller FSM



Short release before HOLDTICKS → one-cycle Press;

Long press $\geq$ HOLDTICKS → Hold while held, no release Press;

Idle low for RECALLTICKS → Recall remains through next press and clears on release.

Simulation Waveform Snapshot

FSM diagram and waveform plot generated from the implemented design and real simulation data.

Headlamp Button waveform, HOLDTICKS=4 RECALLTICKS=10

Rendered from a Questa VCD export. Light dotted guides mark every clk rising edge.

